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In [ ]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler, StandardScaler

df = pd.read_csv("/content/test.csv")

print("Initial Data Overview")
print(df.info())

print("\nFirst 5 Rows")
print(df.head())

print("\nDataset Shape:", df.shape)

print("\nMissing Values Before Cleaning:")
print(df.isnull().sum())

plt.figure(figsize=(10,6))
sns.heatmap(df.isnull(), cbar=False, cmap="Blues")
plt.title("Missing Values Heatmap")
plt.show()

df["Age"] = df["Age"].fillna(df["Age"].median())

df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])

df.drop(columns=["Cabin"], inplace=True)

initial_rows = df.shape[0]

df.drop_duplicates(inplace=True)

print("\nDuplicate Rows Removed:", initial_rows - df.shape[0])

df["Pclass"] = df["Pclass"].astype("category")

categorical_columns = ["Sex", "Embarked"]

for col in categorical_columns:
    df[col] = df[col].str.strip().str.capitalize()

minmax = MinMaxScaler()

minmax_columns = ["Age", "Fare"]
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df[minmax_columns] = minmax.fit_transform(df[minmax_columns])

standard = StandardScaler()

df[["SibSp", "Parch"]] = standard.fit_transform(df[["SibSp", "Parch"]])

print("\nCleaned Dataset Information")
print(df.info())

print("\nFirst 5 Rows After Cleaning")
print(df.head())

print("\nMissing Values After Cleaning")
print(df.isnull().sum())

print("\nStatistical Summary")
print(df.describe(include="all"))

```

Initial Data Overview

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 418 entries, 0 to 417

Data columns (total 11 columns):

#	Column	Non-Null Count	Dtype
0	PassengerId	418 non-null	int64
1	Pclass	418 non-null	int64
2	Name	418 non-null	object
3	Sex	418 non-null	object
4	Age	332 non-null	float64
5	SibSp	418 non-null	int64
6	Parch	418 non-null	int64
7	Ticket	418 non-null	object
8	Fare	417 non-null	float64
9	Cabin	91 non-null	object
10	Embarked	418 non-null	object

dtypes: float64(2), int64(4), object(5)

memory usage: 36.1+ KB

None

First 5 Rows

	PassengerId	Pclass	Name	Sex	
0	892	3	James Kelly, Mr.	male	
1	893	3	Needs) Wilkes, Mrs. James (Ellen	female	
2	894	2	Francis Myles, Mr. Thomas	male	
3	895	3	Albert Wirz, Mr.	male	
4	896	3	Lindqvist) Hirvonen, Mrs. Alexander (Helga E	female	

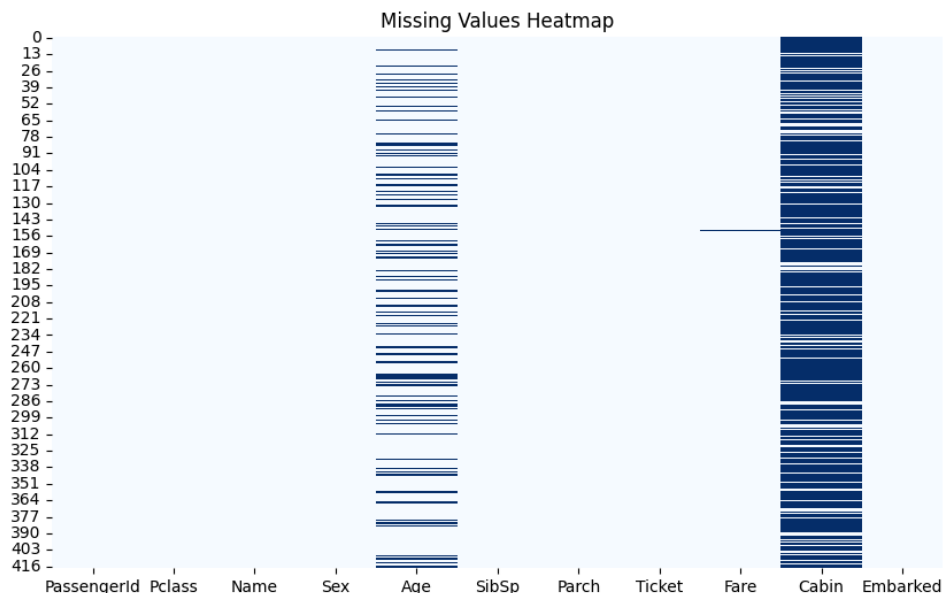
	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	34.5	0	0	330911	7.8292	NaN	Q

1	47.0	1	0	363272	7.0000	NaN	S
2	62.0	0	0	240276	9.6875	NaN	Q
3	27.0	0	0	315154	8.6625	NaN	S
4	22.0	1	1	3101298	12.2875	NaN	S

Dataset Shape: (418, 11)

Missing Values Before Cleaning:

```
PassengerId      0
Pclass           0
Name             0
Sex              0
Age             86
SibSp           0
Parch           0
Ticket           0
Fare            1
Cabin          327
Embarked         0
dtype: int64
```



Duplicate Rows Removed: 0

Cleaned Dataset Information

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 418 entries, 0 to 417

Data columns (total 10 columns):

#	Column	Non-Null Count	Dtype
0	PassengerId	418 non-null	int64
1	Pclass	418 non-null	category
2	Name	418 non-null	object
3	Sex	418 non-null	object
4	Age	418 non-null	float64
5	SibSp	418 non-null	float64
6	Parch	418 non-null	float64
7	Ticket	418 non-null	object
8	Fare	417 non-null	float64

```

9   Embarked    418 non-null    object
dtypes: category(1), float64(4), int64(1), object(4)
memory usage: 30.1+ KB
None

```

First 5 Rows After Cleaning

```

      PassengerId Pclass
Name    Sex  \
0      892      3      Kelly, Mr.
James   Male
1      893      3      Wilkes, Mrs. James (Ellen
Needs)  Female
2      894      2      Myles, Mr. Thomas
Francis  Male
3      895      3      Wirz, Mr.
Albert   Male
4      896      3  Hirvonen, Mrs. Alexander (Helga E
Lindqvist) Female

```

```

      Age    SibSp    Parch    Ticket    Fare Embarked
0  0.452723 -0.499470 -0.400248  330911  0.015282      Q
1  0.617566  0.616992 -0.400248  363272  0.013663      S
2  0.815377 -0.499470 -0.400248  240276  0.018909      Q
3  0.353818 -0.499470 -0.400248  315154  0.016908      S
4  0.287881  0.616992  0.619896  3101298  0.023984      S

```

Missing Values After Cleaning

```

PassengerId    0
Pclass          0
Name            0
Sex             0
Age             0
SibSp           0
Parch           0
Ticket          0
Fare            1
Embarked        0
dtype: int64

```

Statistical Summary

```

      PassengerId Pclass      Name    Sex
Age  \
count    418.000000    418.0      418    418
418.000000
unique      NaN      3.0      418      2
NaN
top      NaN      3.0  Peter, Master. Michael J  Male
NaN
freq      NaN    218.0          1    266
NaN
mean    1100.500000      NaN      NaN      NaN
0.388096
std     120.810458      NaN      NaN      NaN
0.167530
min      892.000000      NaN      NaN      NaN
0.000000
25%     996.250000      NaN      NaN      NaN

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0.301068				
50%	1100.500000	NaN	NaN	NaN
0.353818				
75%	1204.750000	NaN	NaN	NaN
0.469207				
max	1309.000000	NaN	NaN	NaN
1.000000				

	SibSp	Parch	Ticket	Fare	Embarked
count	4.180000e+02	4.180000e+02	418	417.000000	418
unique	NaN	NaN	363	NaN	3
top	NaN	NaN	PC 17608	NaN	S
freq	NaN	NaN	5	NaN	270
mean	-1.593622e-17	-2.974760e-17	NaN	0.069540	NaN
std	1.001198e+00	1.001198e+00	NaN	0.109124	NaN
min	-4.994700e-01	-4.002477e-01	NaN	0.000000	NaN
25%	-4.994700e-01	-4.002477e-01	NaN	0.015412	NaN
50%	-4.994700e-01	-4.002477e-01	NaN	0.028213	NaN
75%	6.169924e-01	-4.002477e-01	NaN	0.061484	NaN
max	8.432229e+00	8.781044e+00	NaN	1.000000	NaN